

Outcomes-based Teaching and Learning (OBTL) Course Outline Template

Academic Year	2025	Trimester/Term/Segment	1
Course Coordinator	Chin Chee Kai		
Course Code	AN6201		
Course Title	Data Analytics Practicum		
Pre-requisites	NIL		
No of AUs	3		
Contact Hours	<u>12</u> weeks full-time		

Course Aims

The Data Analytics Practicum (DAP) aims to tap on industry and business environments as teaching platforms for students to both acquire real-life working experience on applications of data analytics, and also apply theory and skills they have learned from classroom settings to actual project environments. It is the course where theory meets practice. The integral benefits from such real-life environment include extending students' classroom learnings and also serving as a transitioning bridge for MSBA students into business environments involving data science, big data, statistics and machine learning.

Students will work full-time for at least 12 weeks (Trimester) on relevant data analytics/ AI projects in one of the following tracks:

1. **Work Track:** each student will apply to preferred Singapore-registered companies, (for-profit or non-profit) organizations, and agencies for internship practicum positions, possibly with the help of GSCDO (Graduate Study Career Development Office). The practicum positions may be paid, or unpaid, but must be full-time covering the 12-week trimester duration for this course, and endorsed by a company's signed contract offer. Under guidance primarily by company's supervisor, students are expected to be driven to deliver set goals, agreed project scope, and/or related project activities, depending on company's requirements and job description. Each student will also be assigned to a NBS supervisor for milestone guidance and general academic supervision.
2. **Project Track:** for students who are unable to find suitable practicum position in time for the trimester, they will be assigned to faculty supervisors for supervised work on data-analytics-related projects. The supervisor will provide a "company" context and will act both as day-to-day project supervisor as well as NBS supervisor for academic supervision.

Students would work in a group of up to 4 (depending on supervisor's final allocation) on projects delegated by supervisor. These projects may include:

- (a) supervisor's consulting project sponsored by industry partners, or

- (b) supervisor's on-going research project, or
- (c) a capstone project proposed primarily by students.

Each student is expected to be self-driven in the following: reading up plenty of **research materials and papers**, researching on latest developments, performing detailed analyses of collected data, creating analytics or AI model, coding program development and deriving realistic recommendations in the final project paper. While the group of students will deliver a **joint Final Report** (different from project paper) for submission, separate grades are assigned by supervisor to individual student depending on performance, contribution, attitude and outcomes.

Intended Learning Outcomes (ILO)

By the end of the course, you should be able to:

1. Acquire the skill to plan, organize for, and deliver on business analytics or AI solution in a real business organizational setting
2. Structure analysis and recommendations into data analytics or AI deliverables
3. Develop communication skills in acquiring business/ customer requirements and delivering data analytics/ AI results to business users.

Course Content

The content of this course will be the full-time work experience gained from working in a company. Depending on job scope, company and project nature, content may include:

1. Applications of technical data analytics knowledge
2. Learning DA-related software applications and usage in practice
3. Interacting with non-DA colleagues, users and management
4. Translating user requirements to technical applications
5. Translating DA outcomes into management insights
6. Personal growth and appreciation of how DA/AI plays important roles in business decisions
7. Communication skills
8. Other soft skills in business settings, such as presentation, initiatives, reporting, etc.

Assessment (includes both continuous and summative assessment)

Component	ILO Tested	Weighting	Team/ Individual	Assessment Rubrics
1. Pre-course Participation	---	10%	Individual	Appendix 1
2. Course Participation	1, 2, 3	20%	Individual	Appendix 2
3. Meetings	1, 2, 3	30%	Individual	Appendix 3
4. Final Report	3	40%	Individual / Group	Appendix 4
Total		100%		

1. Pre-course Participation:

This refers to your effort, ability, outcome of trying to secure a company internship contract from Trimester 1 to before DAP practicum starts. It also includes attendance to any related mandatory briefing, workshops, events as announced from time to time before actual DAP begins. It is your responsibility and to your advantage to apply to various practicum or internship positions conscientiously and to secure a position as early as possible.

2. Course Participation:

This refers to your effort to deliver and complete the practicum without reneging nor adverse events. This runs for the entire trimester of DAP.

3. Meetings:

This refers to your performance during scheduled meetings with NBS supervisors. Attendance is extremely important. As DAP is, by priority, an AU-bearing academic course, work-related excuses or personal schedule conflicts are not sufficient reasons to be absent from such academic meetings with NBS supervisors.

4. Final Report:

This refers your written submission of self-reflection throughout the DAP journey. This Final Report must not include any company-confidential or technically-sensitive details, but would only be your account of **your learning journey and personal growth**.

Formative feedback

You will receive verbal feedback during Meetings. Should you have email queries outside Meeting sessions, you will receive written feedback on your queries from your assigned NBS supervisor.

Learning and Teaching Approach

Approach	How does this approach support you in achieving the learning outcomes?
Company-based working environment as teaching platform	You will apply theories and skills learned from classroom to real-life business environments, thereby gaining first-hand knowledge about conditions, limitations, adaptations, approximations of theories when they are applied. You will further acquire necessary communication skills to understand user requirements and present your analysis in ways that non-data-analytics colleagues could understand.

Reading and References

Suggested Online References

(PT) Python 3 Tutorial: <https://docs.python.org/3/tutorial/>
(SQ) SQLite Library: <https://www.sqlite.org/index.html>
(PL) Pandas Library: <https://pandas.pydata.org/>
(MP) Matplotlib Library: <https://matplotlib.org/>
(TFT) TensorFlow Tutorial: <https://www.tensorflow.org/tutorials/>
(KT) Keras Tutorial: <https://keras.io/>
(HT) Hadoop Tutorial: <https://www.tutorialspoint.com/hadoop/index.htm>
(ST) Spark Tutorial: <https://spark.apache.org/docs/latest/quick-start.html>

Other Reference Textbooks

(WP) William F. Punch, The Practice of Computing Using Python, 3rd Edition, 2017, Pearson, ISBN 978- 1-2921-6668-1.
(ML) Mark Lutz, Learning Python, 5th Edition, 2013, O'Reilly Media, ISBN 978-1-4493-5573-9.
(WW) William Wesley McKinney, Python for Data Analysis, 2nd edition, 2017, O'Reilly Media, ISBN 978-1-4919-5766-0
(AB) Avrim Blum, Foundation of Data Science, 2015
(PH) Patrick Henry Winston, Artificial Intelligence, 3rd Edition, Addison-Wesley Publishing Company, ISBN 0-201-53377-4.
(SR) Sebastian Raschka, Python Machine Learning, 2017, 2nd edition, Packt Publishing, ISBN 978-1787125933.
(SB) Steven Bird, Ewan Klein & Edward Loper, Natural Language Processing with Python, 3rd edition, 2009, O'Reilly, ISBN 978-0596516499.

Course Policies and Student Responsibilities

(1) General

You are expected to complete all academic requirements punctually and conscientiously by due dates. You are expected to take responsibility to actively apply for practicum internship opportunities from Trimester 1 onwards. The DAP is an AU-bearing academic course. As such, you should prioritize completion of DAP and seek to be flexible, adaptive and explorative in securing practicum opportunities, instead of imposing high personal expectations such as demanding high pay, limiting to only a few companies by industry, by size, by country of origin, by brand, by career goals, by personal favorites, etc. With fixed starting and ending dates for the full-time DAP to be completed, restrictive personal expectations and filtering of opportunities would only lead to inability to begin DAP on-time. Instead, you may seek to try out DAP practicum positions in new industries or new roles different from your past working experience to both explore new job scopes and also expand DAP opportunities to accelerate securing of your practicum position in time to begin DAP.

Unless there are special involuntary circumstances or approved exception by School or University, submission deadlines for all assignments and required submissions will not be extended under all circumstances. If working in a group setting, you are also expected to contribute actively and avoid becoming a burden to your group.

You must complete the 12-week full-time work in the same track and same company (or same supervisor if on Project Track). If both you and company prefer, you may apply to NBS to continue with the company under optional internship extension after completing DAP. As working full-time in the same environment continuously for 12 weeks is an important condition and necessary completion requirement for this course, thus depending on circumstances, you would get Incomplete or Failure grade if there is an interruption (whether due to termination, LOA, switching of tracks, etc) in the 12-week full-time work, and would have to repeat the DAP again in the following year (and follow any new requirement, condition expected of students then).

(2) Good Behavior and Upkeeping NBS/NTU Image

You must at all times exhibit good behavior in upkeeping NBS/NTU image. This includes but is not limited to:

- being punctual, courteous and thankful when attending interviews,
- not renege on signed contract which you have signed and agreed (for better pay, for changing of track, or for any other personal preference or reason), whether it is before or during the internship,
- not terminating or causing or negotiating to be terminated during the internship,
- being positive, constructive and attentive in your day-to-day dealings with company staff and colleagues,

(3) Absenteeism

Absence from Meetings without a valid reason will severely affect your overall course grade. Valid reasons include falling sick supported by a medical certificate and participation in NTU's approved activities supported by an excuse letter from the relevant bodies. Note that attending Meeting is an academic requirement – thus, work-related excuses (company meeting, overtime work, “busy”) or personal schedule conflict (eg, family visits, personal overseas tour, etc) are not valid excuses to skip Meetings.

If you ultimately would miss a Meeting, it is your responsibility to inform your supervisor via email prior to the Meeting and to arrange for a make-up Meeting at the supervisor's available schedule.

(4) Copyright Notice

All course materials (including and not limited to lecture slides, handouts, recordings, assessments and assignments), are solely for your own educational purposes at NTU only. All course materials are protected by copyright, trademarks or other rights. All rights, title and interest in the course materials are owned by, licensed to or controlled by the University, unless otherwise expressly stated. The course materials shall not be uploaded, reproduced, distributed, republished or transmitted in any form or by any means, in whole or in part, without written approval from the University. You are also not allowed to take any photograph, video recording, audio recording or other means of capturing images and/or voice of any of the course materials (including and not limited to lectures, tutorials, seminars and workshops) and reproduce, distribute and/or transmit in any form or by any means, in whole or in part, without written permission from the University. Appropriate action(s) will be taken against you (including and not limited to disciplinary proceedings and/or legal action) if you are found to have committed any of the above or infringed copyright.

(5) Confidentiality and Sensitive Information

Company-confidential information or sensitive materials must not be leaked out during Meetings nor included in Final Report. It is important for you to upkeep the integrity of confidentiality and sensitivity of the company information, as the consequence of such leaks and exposure could be detrimental to the company, and may attract legal liabilities to yourself.

(6) Video Recordings and Photo Taking

All Meetings, whether face-to-face or synchronized online, are subject to be video-recorded in whole or in part, and have photos taken. These digital records are necessary for purposes such as attendance, assessment, evidence of efforts, illustration, potential contact tracing, or official procedures required to conduct the class, satisfy School, University or regulatory requirements. By attending Meetings, you submit that you agree to have such video/ photo records being taken.

Academic Integrity

Good academic work depends on honesty and ethical behaviour. The quality of your work as a student relies on adhering to the principles of academic integrity and to the NTU Honour Code, a set of values shared by the whole university community. Truth, Trust and Justice are at the core of NTU's shared values.

As a student, it is important that you recognize your responsibilities in understanding and applying the principles of academic integrity in all the work you do at NTU. Not knowing what is involved in maintaining academic integrity does not excuse academic dishonesty. You need to actively equip yourself with strategies to avoid all forms of academic dishonesty, including plagiarism, academic fraud, collusion and cheating. If you are uncertain of the definitions of any of these terms, you should go to the [academic integrity website](#) for more information. Consult your instructor(s) if you need any clarification about the requirements of academic integrity in the course.

Course Instructors

Course Coordinator (*)/ Instructor	Office Location	Phone	Email
Chin Chee Kai (*)	---	---	cheekai@ntu.edu.sg

Industry Participation

Company Name	Description of involvement (e.g., co-curation of course, speaker or instructor), include no. of course hours if known.	Contact Person	Email
Capitaland, Sanofi, HP, AStar, GIC, Hilton, etc (above are some examples; companies change from trimester to trimester)	Internship sponsor	N.A.	N.A.

Planned Weekly Schedule

Week	Topic	ILO	Remarks
1	Meeting #1	1, 2, 3	
8	Meeting #2	1, 2, 3	
11	Final Report Submission	3	
12	Practicum Completion Form (PCF) signing	-	PCF must be wet-ink signed in-person face-to-face with NBS supervisor.

Appendix 1: Rubric for Pre-course Participation

Pre-course participation runs from Trimester 1 to just before the DAP course begins. It primarily assesses student's ability, effort, momentum, and outcomes of searching for and securing an internship practicum contract with a company. While it is primarily student's responsibility to secure a company internship contract, there are times when an interested student may still not be able to find a contract offer. The student's effort and actions will be taken into account. Inputs from GSCDO and GSO about the student will also be taken into account.

	Standards		
	Below Expectation (1 – 3)	Meet Expectation (4 – 7)	Above Expectation (8 – 10)
Criteria for Pre-course Participation	Puts in little to no effort in search of practicum opportunities, OR Signed and submitted contract, but then reneged to join another company, OR Not heeding advice by GSCDO/ GSO to improve search opportunities, OR Not submitting signed contract before start of practicum, OR submitting signed contract more than 7 days after date of signing on contract, OR Not attending mandatory briefing, workshop, event, OR Not responding to communications (eg emails, calls, reminders) about	Puts in adequate effort in search of practicum opportunities; though not quite successfully, but kept trying, AND Responding somewhat to communications (eg emails, calls, etc) about progress in practicum search, although occasionally responding after long delay or not responding, AND Submitted signed contract to NTULearn within 3 days after accepting offer, the submission timing being significantly before start of practicum.	Puts in strong effort or uses creative methods in search of practicum opportunities, despite getting repeated rejections or non-offers, AND Managed to secure and sign internship offer, AND Responding readily to communications (eg emails, calls, etc) about progress in practicum search, AND Submitted signed contract to NTULearn soonest within 3 days after accepting offer without delays, the submission timing being significantly before start of practicum.

	practicum opportunities, or about information or progress enquiries by AN6201 staff or supervisors.		
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Appendix 2: Rubric for Course Participation

Course participation runs for the entire trimester in which DAP course is conducted. It primarily assesses the student's general adherence to delivering the signed contract's scope and good behavior with the sponsoring company.

	Standards		
	Below Expectation (1 – 3)	Meet Expectation (4 – 7)	Above Expectation (8 – 10)
Criteria for Course Participation	Company complains about student (with valid reasons), OR Company initiates contract termination with student due to direct, induced, or intentional fault of student, OR Student initiates contract termination for any reason, OR Exhibiting bad mannerism, behavior or action towards company, supervisor or peers, OR Not responding to communications (eg emails, calls, reminders) by AN6201 staff or supervisors about practicum-related matters.	Puts in adequate effort in delivering and completing the practicum, AND Completes all academic submissions on time, AND Responding readily to communications (eg emails, calls, etc) by AN6201 staff or supervisors about practicum-related matters, although occasionally responding with delay or missing.	Met all requirements in "Meet Expectation" column, AND Wins praises from company staff, management, or supervisor, OR Clear evidence in helping fellow course mates, OR Clear evidence in enhancing MSBA/ NBS relationship with company.

Appendix 3: Meetings

Meetings' marks are primarily determined by student's attendance and participation level in meetings with NBS Supervisor.

	Standards		
	Below Expectation (1 – 3)	Meet Expectation (4 – 7)	Above Expectation (8 – 10)
Criteria for Meetings: Attendance Participation	Attendance: very late, absent without informing supervisor, OR absence with informing supervisor with acceptable reason as determined by Supervisor (work, or "busy" are not generally acceptable). Participation: mediocre or brief or reluctant presentation when asked to present. Content: unrelatable, disorganized chain of events and incidents. Shows weak or inability to use those events for introspection. Sounds generic, with broad strokes of "good" or "bad" or without details to give clarity to audience. Overall sounds monotonic, or lacking depth in descriptions.	Attendance: occasionally a little late. Participation: satisfactory presentation when asked to present. Ready to speak when called upon, though sometimes with hesitation, but otherwise would not initiate contribution. Content: rather diary-style event-narration. Does provide a journal of events, achievements, experience, but somewhat missing in ability to pivot those events against some mild level of introspection. Sounds standard, somewhat lacking in practicum-specific details.	Attendance: sustained punctual attendance. Participation: great presentation when called upon, gives inspiring sharing and conversation. Appropriate voluntary contribution to other group members' sharing. Content: organized, relatable, brings out supporting details, perspective-based rather than time-based descriptions, able to show clear self-reflection using events encountered, presented in a way that is inspiring to group members.

Appendix 4: Rubrics for Final Report

Final report is primarily about student's self-reflection about their learning journey throughout the practicum. Final Report will not contain technically sensitive or company-confidential information or references. Work Track students will submit as individual Final Report; an additional project report may be required to be submitted to the company, depending on their requirement. Project Track students will submit as a group one Final Report, in addition to their technical report that may contain technical details which may be sensitive to the project.

	Standards		
	Below Expectation (1 – 3)	Meet Expectation (4 – 7)	Above Expectation (8 – 10)
Criteria for Final Report	Gives plain, predictable description about practicum's work experience that seems to be applicable to any student. Gives vague, little or no link as to how MSBA courses influence the student's growth. Appears to be not aware or sure about how MSBA plays a role in the practicum (whether positively or negatively). Provides generic, little or no detail about report's claim or conclusions, OR Report shows mainly generic statement that appears applicable to any student, lacking individuality.	Gives generic description about practicum's work experience. Gives adequate description or link as to how MSBA courses influence the student's growth. Provides some level of details to characterize the report as the student's experience and work, but could have done more. Overall impression with audience is a well completed practicum experience with some stated personal growth. Though some occasional diary-style narration might be shown, there are signs of ability to	Gives convincing self-reflection of practicum's work experience and knowledge gain, presented in a manner that is inspiring to audience, or future employers. Gives clear link to show how MSBA program or courses influence the student's growth or progress in understanding of real-world challenges in data analytics. Provides unambiguous details and uniquely individual examples, illustrations and/ or situational cases to back report's claim and conclusions. Overall impression with audience is an

	Overall impression with audience is a practicum experience presented like a diary of historical evidence. Even while the diary might be written well, there is little sign of ability to absorb and digest the people, events, and incidents met during practicum for self-reflection – both the “self” and/or “reflection” are missing in major ways in the report.	absorb and digest the people, events, and incidents met during practicum for self-reflection, although either the “self” or “reflection” could be somewhat missing or not adequately illustrated in report.	inspiring practicum experience with deep, honest soul-searching self-reflection, showing clear ability to anchor people, events, and incidents met during practicum into nutrients for self-growth – that the “self” and “reflection” are clearly and undeniably demonstrated in report.
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